

AIM: Generate minimum 10 passwords of length 12 characters using openssl command

Generating Random Passwords with OpenSSL

Concept: Using OpenSSL's rand command to create cryptographically secure random strings.

Command:

```
for i in {1..10}; do  
    openssl rand -base64 12  
done
```

Explanation:

- for i in {1..10} → Loops 10 times.
- openssl rand → Generates random bytes.
- -base64 → Encodes in Base64 (makes it keyboard-friendly).
- 12 → Number of random bytes (Base64 output will be longer than 12 characters).

Sample Output:

```
gqK7F2cYvVZ4rV+2  
zO7S9pHnA4q2K7yG  
N6+Z3K0oVhZrC5lY  
2u8+qEoV1zCk6iGb  
qL5vZ2CkA0XhQ8yP  
rCzG5lN8wO0hK7vQ  
W2+oQpV3lK6jY7aG  
F3xK9uQvP1wH4lZr  
gQ9yV0uP8hZrB7kT  
lZpK2hV6qR8yJ4nC
```

If you need exactly 12 visible characters (no extra Base64 padding):

You can trim the output using cut:

```
for i in {1..10}; do
  openssl rand -base64 12 | cut -c1-12
done
```

Sample Output:

```
gqK7F2cYvVZ4
zO7S9pHnA4q2
N6+Z3K0oVhZr
2u8+qEoV1zCk
qL5vZ2CkA0Xh
rCzG5lN8wO0h
W2+oQpV3lK6j
F3xK9uQvP1wH
gQ9yV0uP8hZr
lZpK2hV6qR8y
```

That one is a **single command** — it just looks long because it's written across multiple lines for readability.

In reality, you can paste it **exactly like this** in your terminal:

```
for i in {1..10}; do openssl rand -base64 12 | cut -c1-12; done
```

This is **not** a separate script file — it's just a **one-liner loop command** in bash that:

1. Loops from 1 to 10 (for i in {1..10})
2. Runs openssl rand -base64 12 (generate random bytes, then encode in base64)
3. Cuts the result to 12 characters (cut -c1-12)
4. Prints each password